Agent Behaviors and Reflexes in Agent-Based Systems

Sombrano, Mrincess Allisha C.
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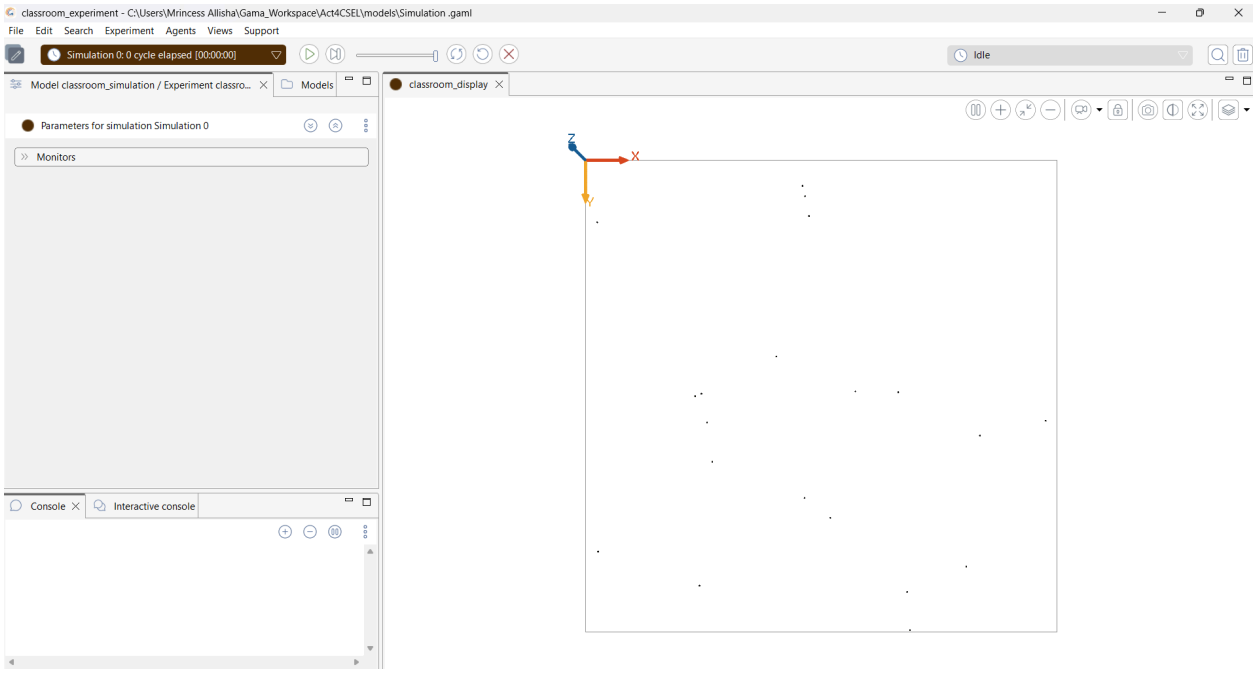
Sir. Mark Bernardino

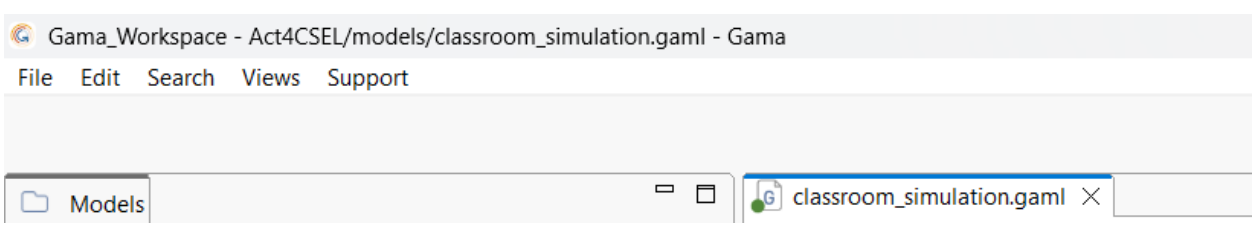
Part 1 – Simulation Scenario

You will create a simple classroom participation simulation.



Part 2 – Step 1: Create the Model

Create a new GAMA model.
Example: model classroom_simulation



Part 3 – Step 2: Define the Student Agent
Create a student species.

```
species student {

    int energy <- 5;
    int score <- 0;
    string status <- "active";
```

Part 4 – Step 3: Add Behavior (Participation)
Students randomly participate in class.

```
    reflex participate when: status = "active" {
        if flip(0.4) {
            score <- score + 1;
            energy <- energy - 1;
        }
    }

```

Part 5 – Step 4: Add Reflex for Status Update
When energy becomes 0, change the status.

```
    reflex update_status {
        if energy <= 0 {
            status <- "inactive";
        }
    }
}

```

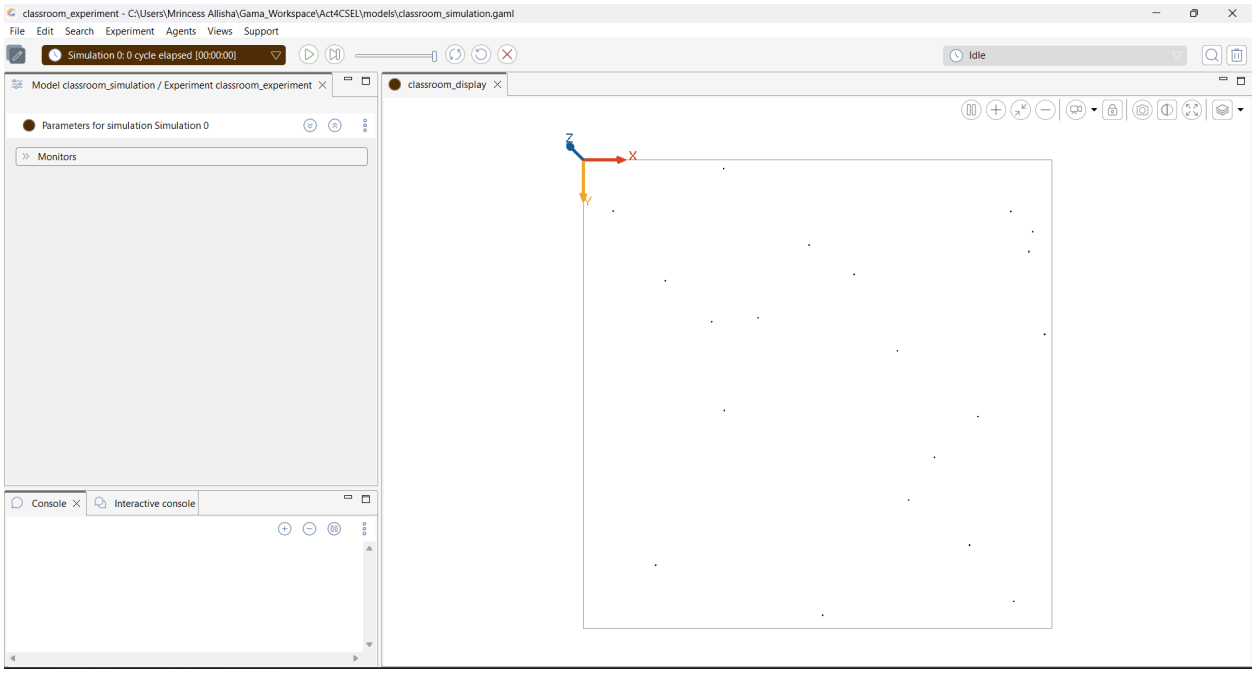
Part 6 – Step 5: Create the Environment
Add the global section.

```
global {

    init {
        create student number: 20;
    }

}
```

Part 7 – Step 6: Run the Simulation



Observe the following:

- **Which students participate the most**
- **How energy changes over time**
- **When students become inactive**

When running the simulation, students participate at different rates depending on chance. Each time a student participates, their score increases by 1 while their energy decreases by 1. Over time, students who participate more often lose energy faster. Once a student's energy reaches 0, they become inactive and stop participating, so their score no longer increases. Observing the simulation shows that some students remain active longer, while others become inactive earlier, and the overall average energy gradually decreases as cycles progress.

Part 8 – Guide Questions

Answer the following questions.

1. **What happens to students when energy reaches 0?**

   When a student's energy reaches 0, their status changes to "inactive." Inactive students can no longer participate in class activities, so their scores stop increasing. Only active students continue to participate and affect the simulation results.

2. **How does participation affect score and energy?**

   Participation increases a student's score by 1 each time, but it also reduces their energy by 1. Students who participate more frequently gain higher scores but deplete their energy faster, while students who participate less often conserve energy but earn points more slowly. This creates a trade-off between performance and stamina.

3. **If participation probability increases to 0.8, what happens?**

   If the participation probability rises to 0.8, students participate much more often. This makes their scores increase faster, but energy also drops more quickly, causing students to become inactive sooner. Higher participation probability speeds up both score gain and energy depletion, leading to earlier "burnout" for many students.

4. **What pattern do you observe in the simulation?**

   The simulation shows that early cycles have most students active, with scores increasing and energy decreasing gradually. In middle cycles, some students start becoming inactive as energy hits 0. By later cycles, only a few students remain active, and average energy and participation decrease. Overall, the pattern illustrates that higher participation leads to higher scores but faster exhaustion.